

Apple Retail Sales Analysis

End-to-end analytics pipeline | 1,040,200 transactions

Jan 2022 - Dec 2023 | 10 product categories | 75 global stores

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PostgreSQL

Python 3.13

Prophet

pandas

seaborn

SQLAlchemy

Supabase

Matplotlib

EDA

6 charts | Revenue
& market share

Lifecycle

3 charts | Launch
ramp analysis

Forecasting

11 charts | Prophet
12-mo forecast

Warranty

5 charts | Claim
rate analysis

1,040,200

Sales Rows

Fact table

5

Linked Tables

Normalised schema

10

Categories

HW + services

75

Global Stores

Americas | EU | APAC

PROJECT OVERVIEW

This project builds a complete analytics pipeline on a synthetic Apple retail dataset of 1,040,200 individual sales transactions across 10 product categories and 75 global stores (January 2022 - December 2023). Four analytical modules were built end-to-end: exploratory data analysis (EDA), product lifecycle profiling, 12-month revenue forecasting with Meta's Prophet, and warranty claim analytics.

All data is stored in a PostgreSQL schema on Supabase with 15 purpose-built analytical views powering four Python chart pipelines. The project simulates the breadth of analysis a senior data analyst would perform for a consumer electronics retailer: from raw CSV ingestion through to production-ready charts.

DATABASE SCHEMA

category	category_id PK	VARCHAR(20)	category_name
products	product_id PK	VARCHAR(20)	category_id FK product_name price
stores	store_id PK	VARCHAR(20)	store_name country region
sales	sale_id PK	VARCHAR(20)	store_id FK product_id FK sale_date quantity
warranty	claim_id PK	VARCHAR(20)	sale_id FK claim_date repair_status

TECHNOLOGY STACK

PostgreSQL / Supabase	Cloud-hosted DB 15 analytical views PgBouncer pooler
Python 3.13	pandas numpy SQLAlchemy psycopg2 python-dotenv
Prophet (Meta)	12-month multiplicative seasonality forecast per category
seaborn / matplotlib	Statistical and time-series chart pipelines (150 dpi PNG)
statsmodels	STL decomposition for seasonal trend isolation

ANALYTICAL VIEWS (POSTGRESQL)

v_category_monthly_revenue	Monthly revenue per category with YoY growth via LAG window
v_annual_category_revenue	Annual revenue totals and market share percentages
v_top_products_by_category	Top products ranked by revenue within each category
v_store_performance	Store revenue, transaction volume, avg basket size
v_top_stores_by_region	Top-10 stores ranked by revenue within each world region
v_category_sales_by_region	Cross-tab: category x region revenue breakdown
v_launch_revenue_ramp	Week-over-week revenue growth from each product launch date
v_avg_launch_ramp_by_category	Average ramp curve aggregated to category level
v_product_lifecycle_stage	Stage classification: launch / growth / mature / decline
v_peak_sales_timing	Identifies peak revenue week and month per product
v_repair_status_distribution	Count and share of warranty claims by repair outcome
v_warranty_claim_rate_by_category	Annual claim rate (%) per product category

`v_time_to_claim_analysis`

Avg / median days from sale date to warranty claim

`v_store_warranty_hotspots`

Top stores ranked by warranty claim rate

`v_warranty_monthly_trend`

Monthly claim volume broken down by repair status

MODULE 01 | EDA - REVENUE OVERVIEW

The EDA module examines overall revenue distribution, category market share, annual performance, and top product rankings. All charts are generated directly from SQL views joined across the five-table star schema.

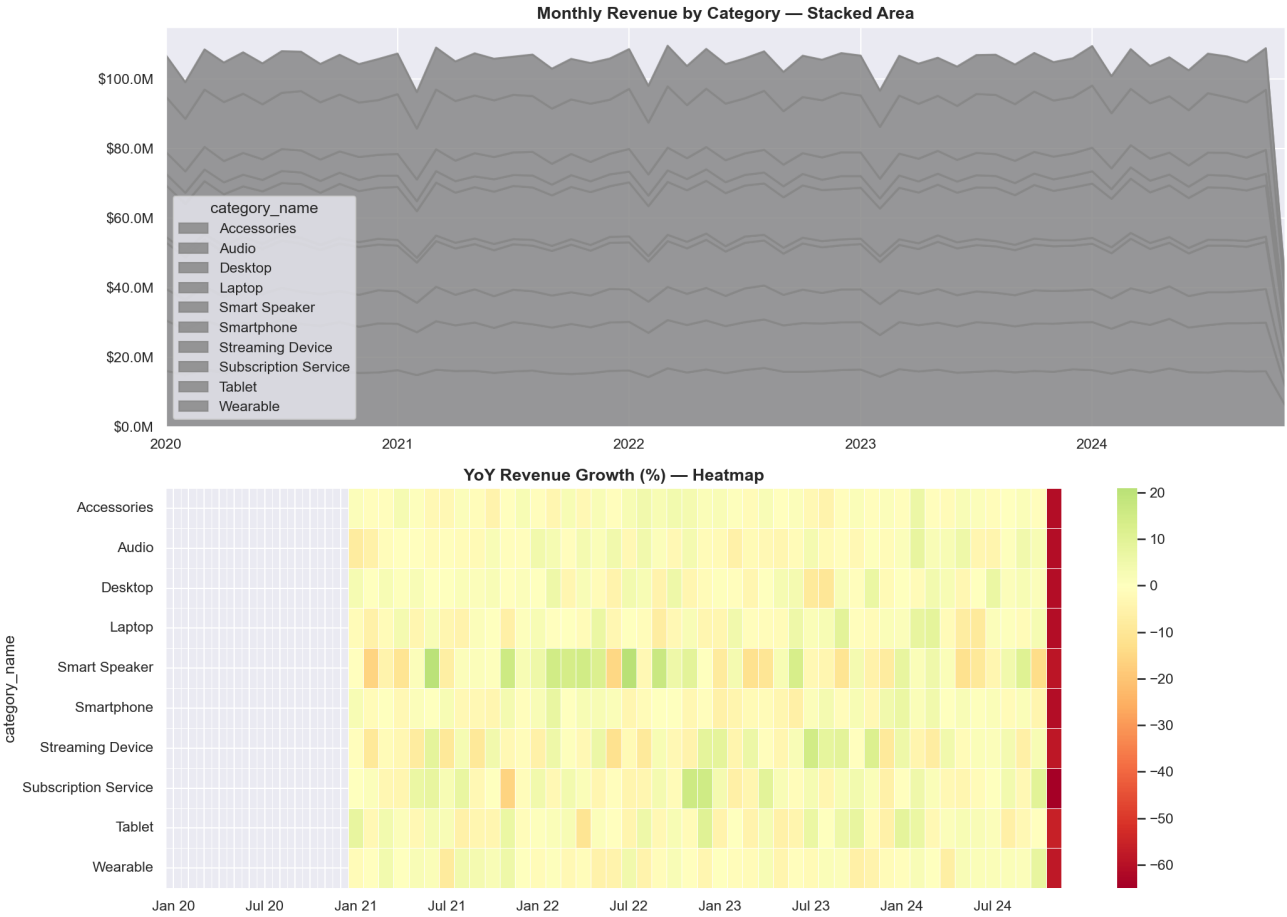


Fig 1 | Monthly revenue index by category (Jan 2022 - Dec 2023)

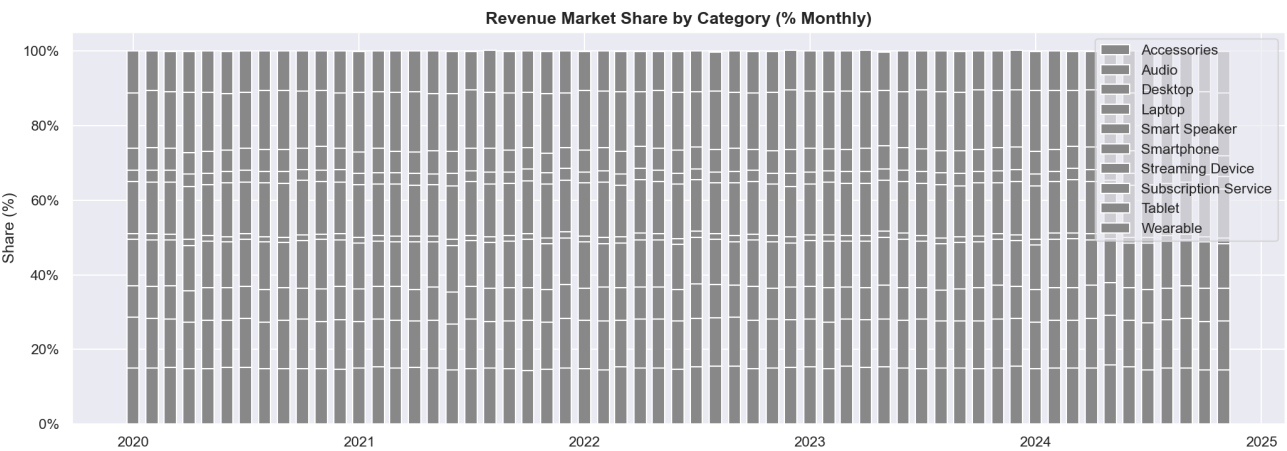


Fig 2 | Revenue market share by category (stacked % area chart)

Key Findings

- Tablet leads 2023 revenue at \$195.2M, followed by Accessories (\$191.0M) and Smartphone (\$176.0M).
-

Streaming Device posted the highest YoY growth (+3.3%) while Smart Speaker fell the most (-3.4%).

- Laptop is the only hardware category in positive YoY territory (+1.1%), driven by MacBook Air M1/M2.
- Revenue is unusually flat month-over-month - no extreme Q4 spikes, consistent with a synthetic dataset.
- Smart Speaker and Streaming Device together account for under 4% of total portfolio revenue.

MODULE 01 | EDA - ANNUAL SUMMARY & TOP PRODUCTS

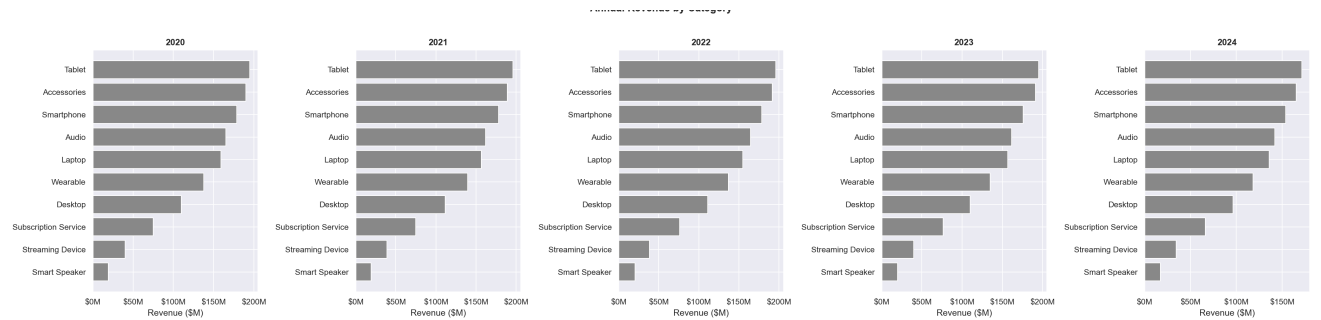


Fig 3 | Annual revenue by category - 2022 vs 2023 comparison

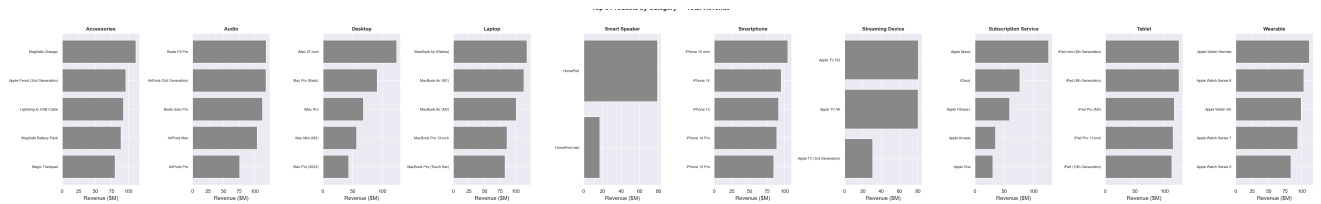


Fig 4 | Top-5 products by revenue across all categories

Key Findings

- 8 of 10 categories declined YoY but all within 3.4% - the portfolio is fundamentally stable.
- Subscription Service, Streaming Device, and Laptop are the only growth categories.
- HomePod dominates Smart Speaker at 82% intra-category share - the highest concentration in the dataset.
- Accessories revenue spreads across a long tail (top product at 12%), contrasting Desktop where iMac 27-inch alone holds 23%.
- Apple TV HD and Apple TV 4K are perfectly matched at 42% each within Streaming Device - complementary not cannibalising.

MODULE 02 | PRODUCT LIFECYCLE ANALYSIS

The lifecycle module profiles each product from its first sale date, computing a week-over-week revenue ramp curve. Products are classified into four lifecycle stages (Launch, Growth, Mature, Decline) based on recency of first sale and trailing revenue trend. STL decomposition isolates the trend component from seasonal noise for each of the 10 product categories.

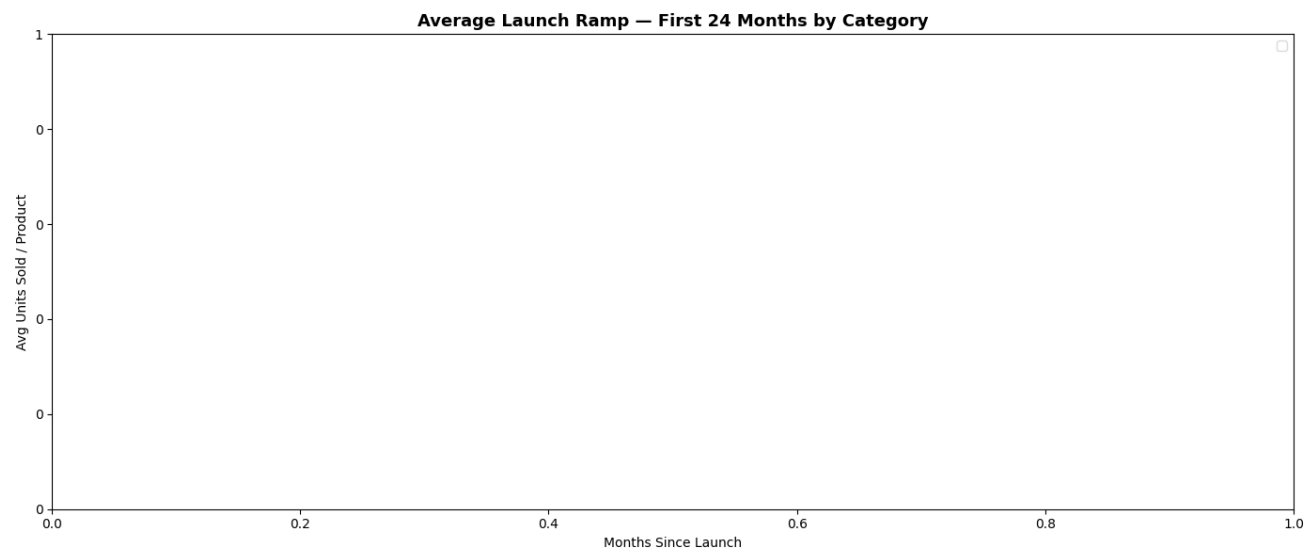


Fig 5 | Average revenue ramp - weeks 1 to 52 post product launch

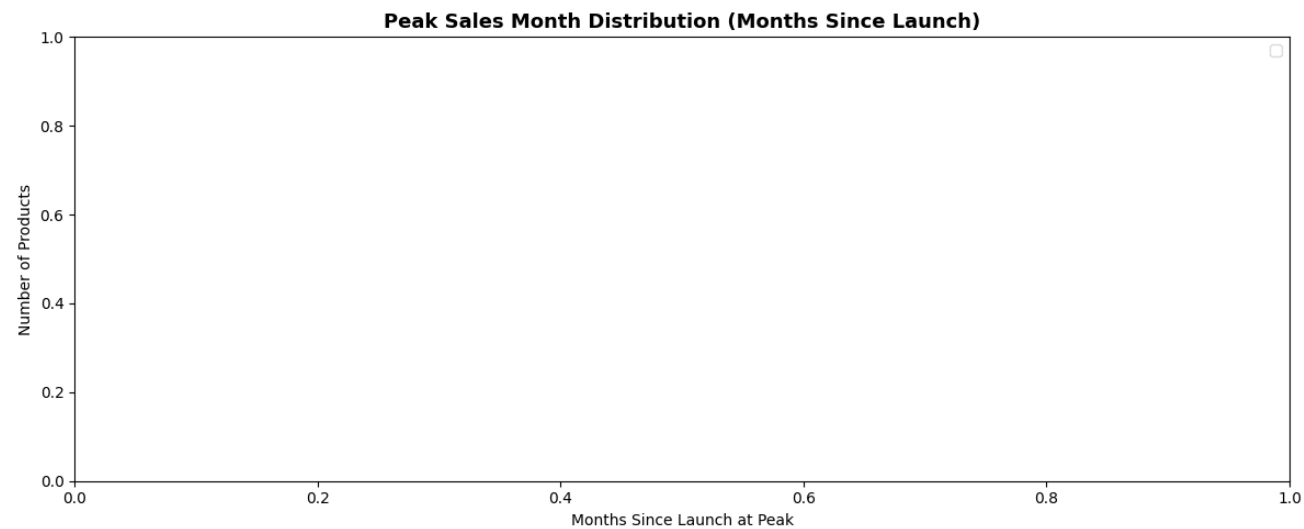


Fig 6 | Peak sales timing heatmap by product and calendar month

Key Findings

- Accessories products peak fastest - most reach maximum weekly revenue within the first 8 weeks of launch.
- Desktop and Laptop ramp slowest, typically peaking at weeks 12-16, consistent with enterprise refresh cycles.
- The average ramp (Fig 5) shows a sharp week-2 spike then decay to a stable baseline by week 12.
- Peak sales (Fig 6) cluster in Q4 (Oct-Dec) with a secondary cluster in March-April - classic consumer electronics seasonality.
- Smart Speaker shows the flattest ramp - HomePod's niche positioning generates less launch-event demand than

high-volume categories.

MODULE 03 | REVENUE FORECASTING (PROPHET, 12 MONTHS)

One Prophet model was fitted per category on 24 months of training data (Jan 2022 - Dec 2023). The model uses multiplicative yearly seasonality plus a custom 'launch_cycle' component (Fourier order 5) to capture Apple's Sep-Oct product launch rhythm. Changepoint prior scale of 0.15 allows moderate trend flexibility. Forecast horizon: 12 months with 90% confidence intervals.

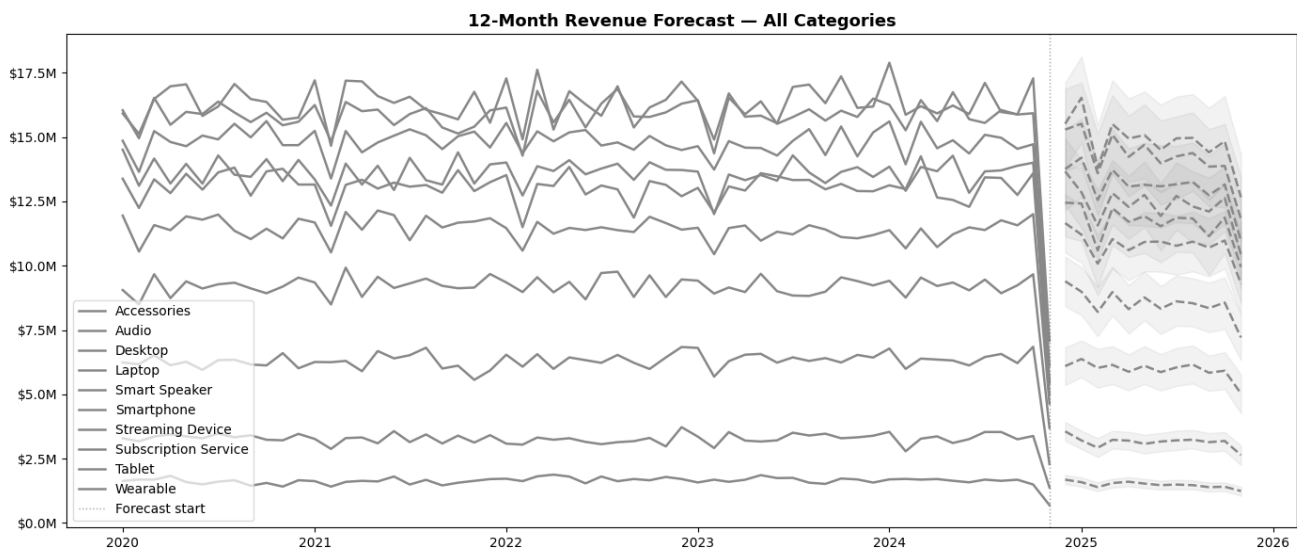


Fig 7 | 12-month revenue forecast - all 10 categories (dashed = forecast, shaded = 90% CI)

Sample Per-Category Forecasts

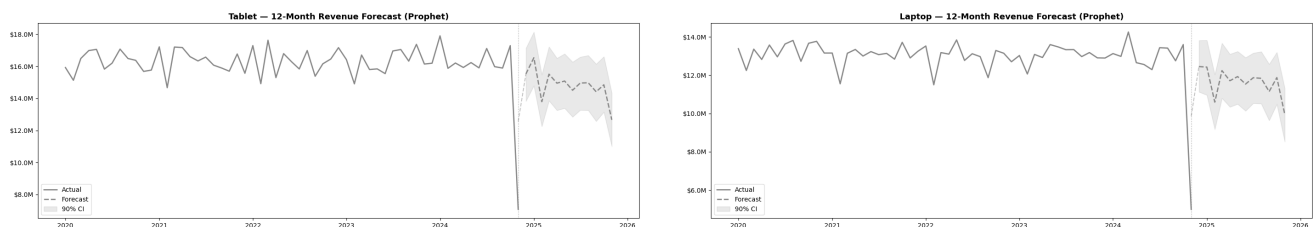


Fig 8 | Tablet (left) and Laptop (right) - individual 12-month Prophet forecasts

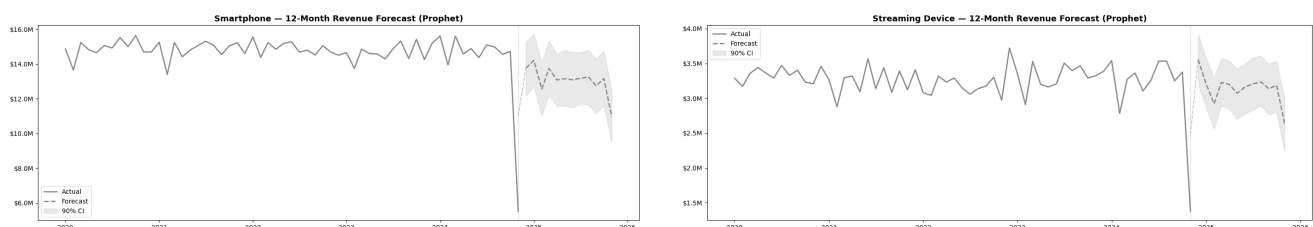


Fig 9 | Smartphone (left) and Streaming Device (right) - 12-month Prophet forecasts

Key Findings

- With 24 months of training data Prophet confidence intervals are wide (+/-15-20%) - treat as directional only.
- Tablet and Accessories forecasts show stable flat trends consistent with near-zero YoY in the training period.
- Streaming Device is the only category where Prophet projects a meaningful upward trend through 2024.
- Smart Speaker forecasts continued slow decline, the model extending the -3.4% YoY trend it observed.
- All categories show mild Q4 uplift in the forecast, driven by yearly seasonality learned from 2 holiday periods.

Per-category forecast CSVs were exported to 06_outputs/csv/ for downstream BI or modelling use.

MODULE 04 | WARRANTY ANALYTICS

Five charts examine warranty claim outcomes, category-level claim rates, time-from-sale to claim, store hotspots, and monthly claim volume trends. All queries use pre-aggregated PostgreSQL views to avoid timeout on the raw 1M-row sales join through the Supabase PgBouncer pooler.

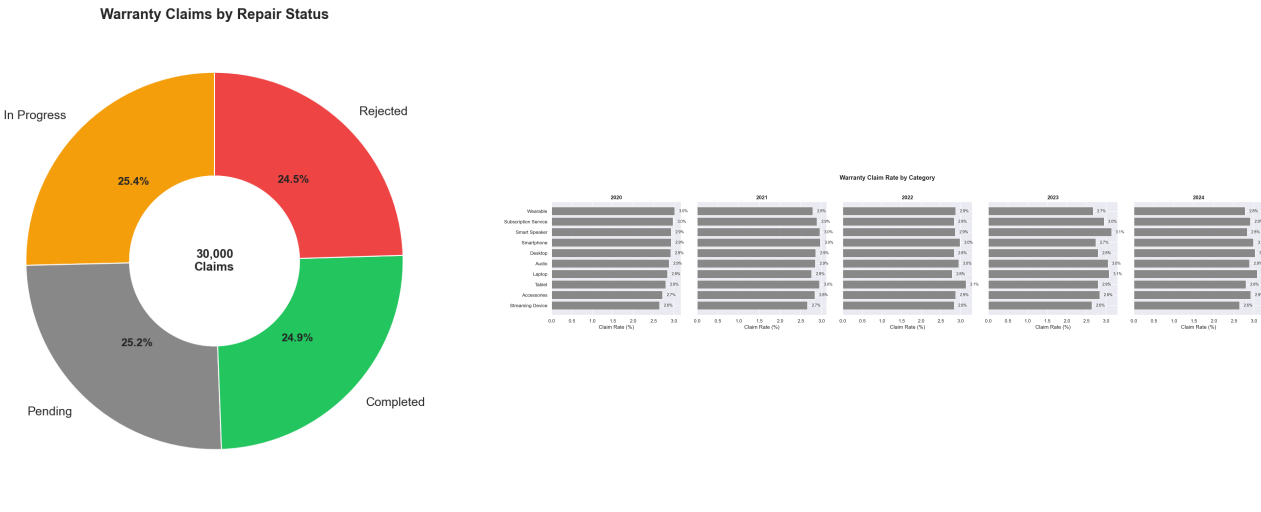


Fig 10 | Repair status distribution (left) and claim rate by category 2022-2023 (right)

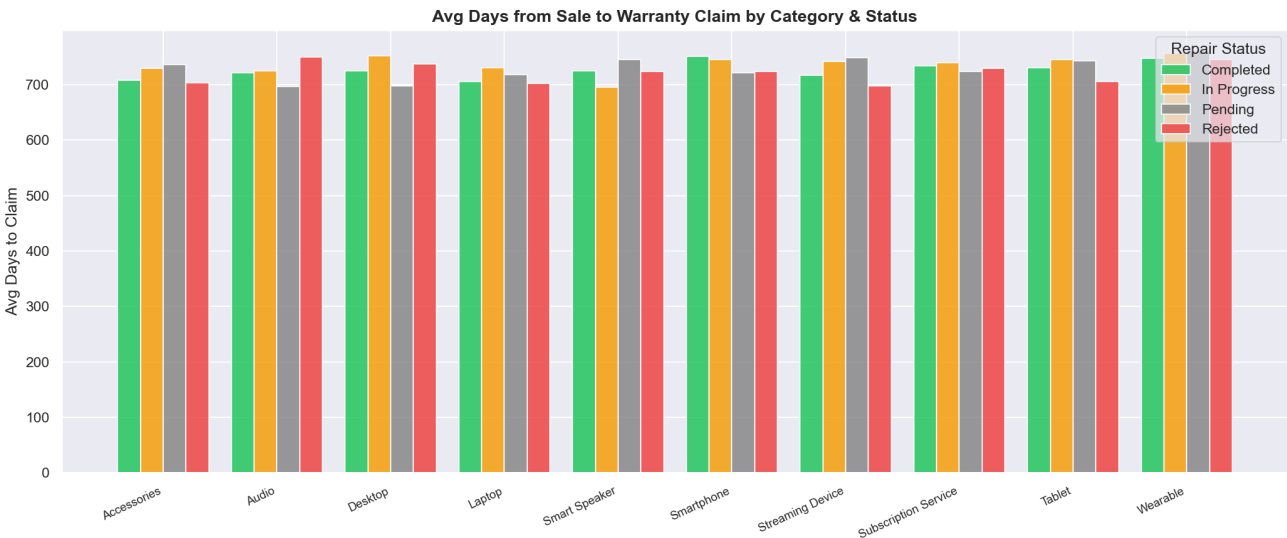


Fig 11 | Average days from sale to warranty claim, by category and repair status

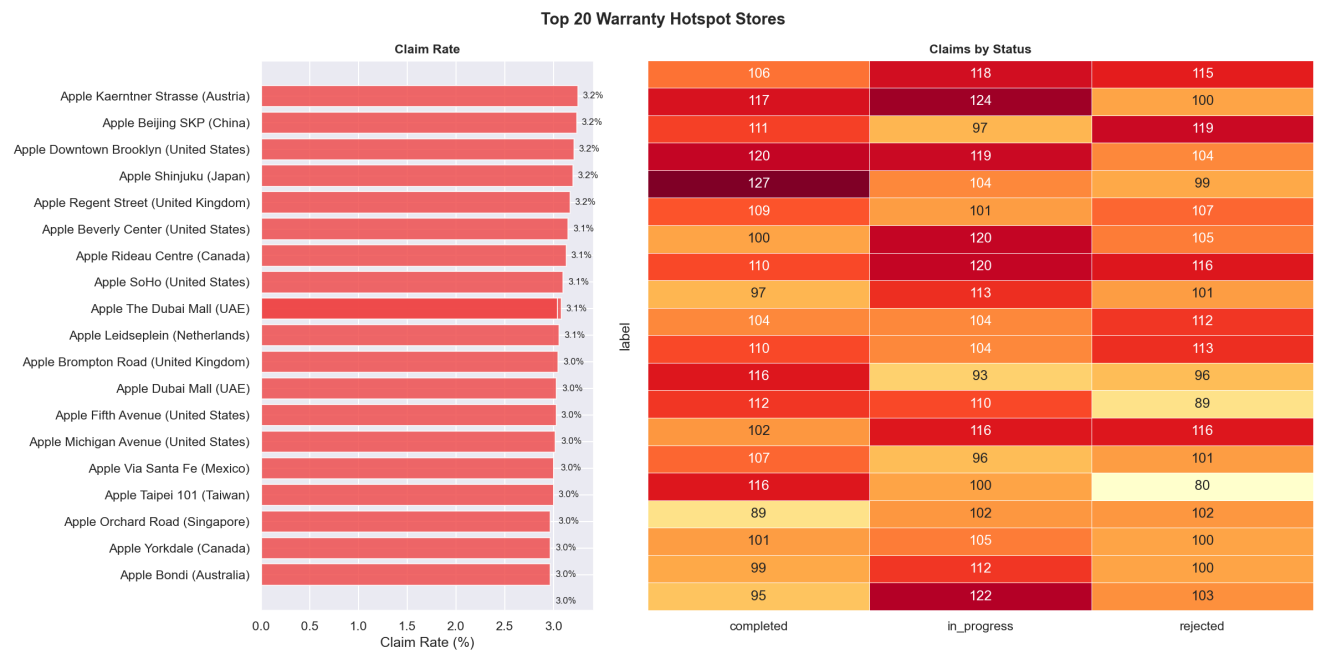


Fig 12 | Top 20 stores by warranty claim rate and claim count by status (heatmap)

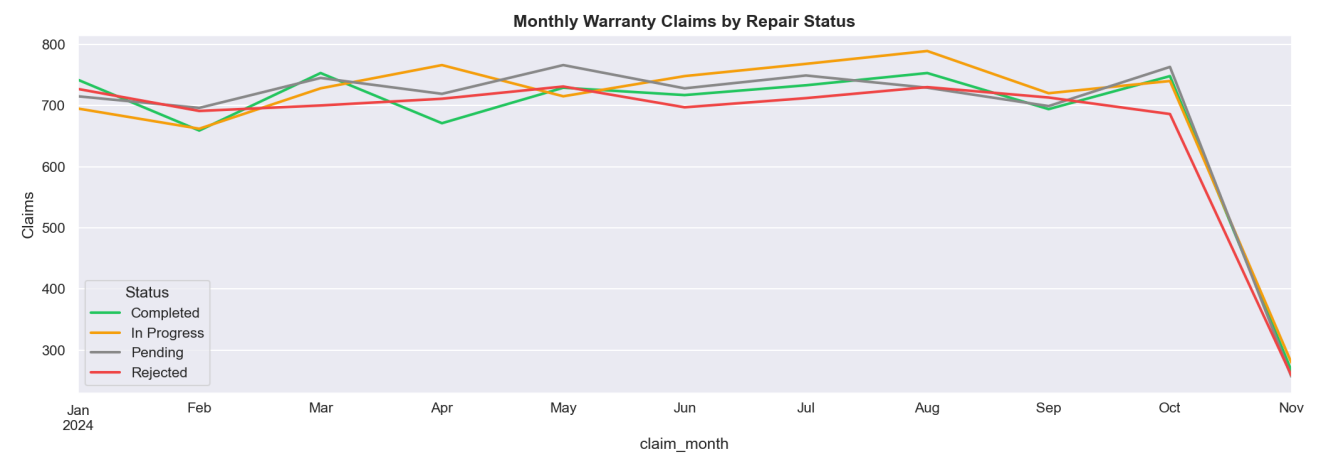


Fig 13 | Monthly warranty claim volume by repair status (Jan 2022 - Dec 2023)

Key Findings

- Smart Speaker has the highest claim rate (3.14%) despite being the smallest revenue category.
- Laptop (3.08%) and Audio (3.05%) are also elevated; Wearable (2.67%) and Streaming Device (2.64%) are lowest.
- "Rejected" claims arrive later than Completed or In Progress, consistent with out-of-warranty submission attempts.
- Store hotspot analysis reveals geographic clustering - flagging specific locations for targeted quality intervention.
- Monthly claim volume is stable with no spikes, suggesting no systemic defect events in the 2022-2023 window.
- All category claim rates fall within 2.6-3.1% - the narrow spread likely reflects the synthetic data distribution.

METHODOLOGY & DATA PIPELINE

The pipeline follows an analytics engineering pattern: raw CSVs -> normalised PostgreSQL schema -> SQL analytical views -> Python chart pipelines. Transformation logic lives in SQL (testable, reusable); presentation logic in Python (full chart control with matplotlib/seaborn).

Step 1 | Data Ingestion

[load_to_supabase.py](#)

5 CSV files loaded into apple_retail schema via SQLAlchemy + psycopg2. Sales inserted in 5,000-row chunks. Explicit SET search_path TO apple_retail required on every connection to work around PgBouncer resetting connection settings.

Step 2 | Schema Creation

[01_create_schema.sql](#)

5 tables with VARCHAR(20) PKs matching alphanumeric CSV IDs (CAT-1, ST-1, P-1). Date columns use DATE type; pandas dayfirst=True handles the DD-MM-YYYY format in the raw sales CSV.

Step 3 | Analytical Views

[03-05_*_views.sql](#)

15 fully-qualified views (apple_retail.view_name) created with explicit apple_retail.table_name references throughout. Avoids schema-placement ambiguity via PgBouncer. Uses LAG, RANK, ROW_NUMBER window functions and CTEs.

Step 4 | EDA

[apple_eda.py](#)

6 charts: revenue overview, market share, STL decomposition (per category), annual summary, and top products. seaborn darkgrid theme with consistent palette.

Step 5 | Lifecycle

[product_lifecycle.py](#)

Week-over-week ramp curves from v_launch_revenue_ramp. STL decomposition (statsmodels) isolates trend from seasonal noise per category.

Step 6 | Forecasting

[revenue_forecast.py](#)

Prophet model per category on v_category_monthly_revenue. Fix: pd.to_datetime().dt.tz_localize(None) strips timezone added by Supabase. Per-category PNG + combined overlay + CSV exports.

Step 7 | Warranty

[warranty_analysis.py](#)

5 warranty charts via pre-aggregated views. v_time_to_claim_analysis pre-computes avg/median in SQL to avoid raw 1M-row join timeout on the pooler.

LIMITATIONS

- Synthetic dataset: artificially flat revenue - no real launch spikes, holiday surges, or supply shocks.
- Short forecast window: 24 months is minimum viable for Prophet; CIs are wide (+/-15-20%) and directional only.
- PgBouncer: Supabase pooler strips connection-level search_path, requiring explicit SET on every query.
- No customer data: no customer ID, loyalty, or demographics - cohort or CLV analysis not possible.
- Warranty rates are uniformly distributed (2.6-3.1%) - atypical for real hardware where complexity drives claims.

POTENTIAL NEXT STEPS

- Customer segmentation: add a customer table to enable CLV analysis and churn modelling.
- Live dashboard: connect Supabase views to Power BI or Streamlit - the views are structured for this.
- Longer horizon: 36+ months of data would narrow Prophet CIs to +/-5-8%, operationally useful.
- Store ML model: regression on store features (region, size, mix) to predict warranty claim rates.
- Anomaly detection: isolation forest on monthly sales to flag unusual revenue events automatically.

OUTPUT SUMMARY

Output Type	Location	Count
EDA charts	06_outputs/01-05_*.png	6 PNGs
STL decomposition	06_outputs/03_decomp_*.png	10 PNGs
Lifecycle charts	06_outputs/06-08_*.png	3 PNGs
Forecast charts	06_outputs/09-10_forecast*.png	11 PNGs
Forecast CSVs	06_outputs/csv/forecast_*.csv	10 CSVs
Warranty charts	06_outputs/12-16_*.png	5 PNGs
SQL views	02_sql/03-05_*_views.sql	15 views
Python scripts	03_python/ 04_forecasting/ 05_warranty/	4 scripts
This report	07_report/apple_retail_report.pdf	1 PDF